(1)

Brightlearn Data Analytics
Name : Cynthia Mmantebaleng Phakela
Course : Data Analytics and Data Science
Assignment No. : Research Assignment 1

Section A : Data Fundamentals

Question 1

Database : Definition and purpose

A database is a structured collection of data that is stored, maintained, and queried to support reliable storage, retrieval and analysis of information. Databases support transactional processing, reporting and decision support by organising data into schemas that enable efficient access and consistancy.

Why are they essential

Modern organisations rely on databases to manage operational data, support analytics and enable timely decision-making. Database systems provide mechanisms for data integrity, transaction processing and queryable access that scale across enterprises; enabling business intelligence, reporting and automated processes.

Real world Examples of where databases are Used

1. E-commerce order processing : An operational

database supports customer orders, inventory, payments and fulfillment. Providing real time or near real time visibility for order status and stock levels. For example Amazon which uses databases to store customer information, order history and product details

② Banking transactional systems: Financial institutions rely on databases to record deposits, withdrawals, transfers and account activities, enabling accurate balances and timely compliance reporting. This is often intergrated with BI to support risk, compliance and analytics.

Question 2

Data Warehouse : Definition and Difference

A data warehouse is a big, central storage place for company's old data. It collects data from many different systems and organises it so it's easy to search and analyse.

Unlike normal databases that handle daily tasks like sales or orders, a data warehouse is built for asking big questions and looking at trends over time. It stores data in a neat way, like using star schemas or data marts, so people can run reports and make smart business decisions

Example

Supermarket Chain: (eg. Pick 'n Pay)
Each store has a database that tracks today's sales and stock. The data warehouse pulls all that sales data from every store for the last 5 years. Managers use it to answer questions like "Which products sell best in winter in Gauteng?" Normal store databases would be too slow for that kind of information extraction over a period.

Why Use a data Warehouse

They are used to enable decision support across the enterprise by integrating data from disparate Source systems, providing consistent metadata and supporting analytical tools such as SQL and OLAP. As per the above example, executives and managers as well as analysts are able to perform trend analysis, forecasting and performance measurement on historical data.

- Data marts relation

Data marts are subject or department specific subsets of a data warehouse. They are designed to serve targeted analytical needs with faster response times and simpler access for particular user groups.

Question 3

Data Lake : Concept

A data lake is one big storage place for all kinds of raw data. It keeps data exactly as it came in ; whether it's tables, text files, videos or sensor data.(stores it as structured, semi-structured or unstructured). It is sorted out later when it is needed for reports, machine learning or other analytical purposes.

Key differences from a data warehouse

- Data Lakes : Store raw data in any format. It is fexible and cheap to store. You decide how to use the data later. It is good for big data and machine learning.

- Data warehouse : Stores clean, structured data that is ready to use. The data is checked and put into neat tables before it is stored. It is fast for reports and business analysis. Data warehouse typically uses schemas like snowflake and ETL (Extra transform load) process to enforce data quality and governance prior to analysis.

When to prefer a data lake

When a business needs to ingest large volumes of diverse data (logs, sensor data, multimedia, social feeds) with low upfront transformation costs; or when raw data should be preserved for exploratory analytics, machine learning and data science workflows. Data Lakes support flexible experimantation and can complement data warehouses by acting as a source of raw data or a staging area for later refinement.

Question 4.

Data Mart : Definition and relation to data warehouse

A data mart is a subset of a data warehouse, focused on a specific business area or department. It is a smaller, more specialised repository that provides fast and easy access from a subset of the warehouse's data. A data mart is typically derived from the data warehouse but can be independent.

Who uses data marts

They are used by department-level analysts and business users who require quick access to a targeted slice of data; eg marketing, finance and sales analytics. They provide faster response times for common queries within a particular business domain.

Example

A big retail company has a data warehouse with housing the whole company's data. eg sales, HR, finance, stock, marketing - all from the past 10 years. The marketing team does not need all the data, so the company creates a Marketing Data Mart for the marketing department.

The data mart (Marketing data mart) has the following information

- customer age, location and gender
- Campaign names and costs
- Ad clicks and sales from campaigns
- email open rates

The marketing team can now quickly check which ad worked best in a specific surburb without having to filter through the whole company's data.

Other examples:

- Sales data Mart: Only holds product sales, targets and store performance. Used by sales managers to analyse data for sale optimisation.
- HR Data Mart: Only holds employee information, salaries and attendance. Used by the HR department.

Question 5

Structured data

Organised data with a predefined format, like tables or spreadsheets. Data is organised in a fixed schema suitable for relational databases and BI tools.

Example:

Excel sheet for employee attendance

Name	Date	Clock in	Clock out
Sarah	14/4/2026	8:05	17:02
James	14/4/2026	8:10	16:58

Semi-Structured data

Semi-structured data doesn't sit in neat rows and columns like a spreadsheet, but it still has labels or tags that help you understand what each piece means. Basically, it is data with tags or markers that provide organisational structure but not a rigid schema (eg JSON - JavaScript Object Notation)

Example

When a weather app asks a server "What's the weather in Alberton?", the server might reply with JSON like:

<> Code

```
{
   "city": "Alberton",
   "temperature": 24,
   "Conditions": "Sunny"
```

Your phone reads that and shows "Alberton: 24°C, Sunny".

Labels are "city", "temperature" and "conditions"

Unstructured Data

It is data that does not follow a fixed format or table. There are no rows, columns or tables. It is just raw content that computers find harder to search and organise without extra work.
Ie: Data without a predefined schema or metadata structure, often binary or free form text/media

Examples

A photo or video : A picture of a dog or a TikTok video : There is no built-in table saying "dog = yes" or "person = smiling". it is just pixels
A word document or PDF report : A 20 page business report has headings and sentences, but the information is not in neat fields like a spreadsheet. You have to read it to understand it

Rules

Structured = Excel table: name, age, city
Semi Structured = Email or JSON : has some labels like subject and Date
Unstructured = Photo, video, audio, freee text: no built in lables at all

Question 6

Data Modeling : What is it and why it matters

Data modeling is the process of defining how data is organised, linked and handled in a system. It produces clear models that show data structures, relationships rules and meanings. These models help design databases, keep data consistent and support accurate analysis and daily operations. Good data modeling is essential for building data systems that can grow, are easy to maintain and match business needs with technical design.

Types of Data Models

① Conceptual Data Model

It is a big picture view that shows the main ideas and how they connect. There are no technical details. It is mainly used by business people and managers.

e.g : For a university

Entities : Student, Course, teacher Relationships : Students enroll in courses. Teachers teach courses

This is just a box and lines. NO field names or data types yet

② Logical Data Model

Adds more detail. Defines all the fields, data types and rules. It still does not say what database software to use. It is mostly used by Data analysts and designers.

eg: same university, but now

Student table : StudentID(number), Name (text), email(text), DateOfBirth(date)
Course table : CourseID (number), CourseName (text), Credits (number)

StudentID must be unique as a student can enroll in many courses

③ Physical Data Model

It is the blueprint for building the database. It includes exact table names, indexes, storage and specific database tech like MySQL or SQL Server. It is used by database developers (DBAs).

eg: same university, now in SQL Server

< > Code

```
CREATE TABLE Students (StudentID INT
 PRIMARY KEY, Name VARCHAR(100) NOT
 NULL, Email VARCHAR(150) UNIQUE, DateOfBirth
 Date
```

It also says where the table is stored, what indexes to add for speed etc

Type	Level	Focus	Example
Conceptual	Big idea	What things exist	Student - enrolls in - course
Logical	Detailed plan	What fields and rules	Student ID, name, Email, must be unique
Physical	Ready to Build	Exact code and storage	CREATE TABLE Students (...) in SQL Server

Question 7

Data Mining Definition and Purpose

Data mining is the process of discovering patterns, relationships and insights from large datasets using statistical, machine learning and database techniques. It supports customer segmentation, prediction, anomaly detection and decision support in business contexts to optimise operations.

Business Uses

- Customer segmantation and targeted marketing
- Fraud detection, risk management and forecasting
- Operational optimisation and anomaly detection in process data

Examples

- Health care research data marts: mining biomedical data in data warehouses and data marts enables discovery of patterns across patient cohorts, supporting clinical studies and translational research pipelines.

- Banks using data marts to detect fraud:
(1) This would include the bank pulling only fraud related data from the main data warehouse into smaller Fraud Data Mart. This includes transations, customer habits, device information and past fraud cases.
Structure is for speed: The data mart uses
(2) Simple layout like star schema so analysts can run fast queries. Example: 'Find all high-value payments outside South Africa in the last 10 minutes.'
(3) Apply detection methods: The fraud team runs rules, pattern checks, and machine learning on the data mart
  - Rules: Flag 3 ATM withdrawals in different cities within 1 hour.
  - Patterns: Spot spending that does not match a customer's normal behavior.
  - ML scores: Rate each transaction for fraud risk.
(4) Take Quick action: If something looks suspicious, the bank can instantly block the card, alert the customer, or send it to an

investigator.

Data marts help as they are smaller, faster and built just for the fraud team. So they can catch fraud in seconds without slowing down the bank's main systems.

Question 8

Database Mangement System (DBMS) : Concept

A DBMS is a software system that provides an interface to store, retrieve, update and manage data in databases. It enforces data integrity, security, concurrency and query processing for various data workloads.

Four Popular DBMS Platforms

① My SQL: is a free, open source relational database system used to store and manage structured data. It organises data into tables with rows and columns and you use SQL (Structured Query Language) to add, find, undate or delete data. It is one of the most popular databases for websites and applications because it is fast, reliable and works well with languages like PHP (Hypertext Preprocessor), Python and Java

Example

A website like an online store uses MySQL to keep tables of Customers, Products and Orders so that it can quickly check stock or pull up your past purchases

- web applications with structured transactional data

(2) PostgreSQL

It is a free, open source relational databes system like MySQL, but with more advanced features. It stores data in tablesusing rows and columns, you use SQL to work with it. it is known for being very reliable and good at handling complex queries, large amounts of data and custom datatypes.

Example:

Many banks, tech companies and apps use PostgreSQL to store customer accounts, transactions or location data because it supports advanced rules and stays accurate even with huge workloads

(3) Oracle

Oracle database is a commercial relational database system built by Oracle Corporation. it stores data in tables and uses SQL, like MySQL and PostgreSQL It is known for handing very large, complex, high security workloads for big companies, banks and governments

Example

A major air line might use Oracle Database to manage millions of bookings, flight schedules and loyalty points worldwide, because it is built for heavy traffic, strong security and almost no downtime

④ MongoDB

It is a free, open source NoSQL database that stores data in flexible documents instead of tables. Instead of rows and columns, it uses JSON like documents. Each document can have different fields, so that there is no need for a fixed structure upfront.

Key points

- Schema-flexible: easy to add new fields anytime
- fast for big, changing data: good for realtime apps, mobile apps, catalogs
- scales out easily: built to spread data across many servers

Example

A social media app might use MongoDB to store user profiles. One user's document can have name, posts, photos. Another can have name, video, location. There is no need to change the whole database structure.

(16)

	MySQL	PostgreSQL	Oracle Database	MongoDB
Type	Relational (SQL)	Relational (SQL)	Relational (SQL)	NoSQL-Document
Cost	Free, open-source	Free, open source	commercial, expensive license	Free, open-source. Paid cloud option
Data format	Tables: rows & columns. Fixed schema	Table: rows & columns. Fixed schema.	Tables: rows & columns. Fixed schema.	JSON-like docs. Flexible schema.
Best for	Websites, apps, read heavy workloads	complex queries analytics, data integrity	huge enterprise systems needing 24/7 uptime	real time apps, fast changing data, mobile apps
Strengths	Easy to use, very fast, huge community	Very advanced SQL features, reliable, handles complex data types	Top tier security clustering, tools for massive scale	Flexible structure easy to scale out, great for JSON
Weakness	Fewer advanced SQL features than PostgreSQL	Can be harder to tune for raw speed vs MySQL	costly, complex. Needs expert DBAs	not great for heavy joins or strict trans-actions
Who uses it	WordPress, YouTube, Shopify	Instagram, Apple, banks	Banks, air lines governments	Uber, Forbes

Question 9

ETL (Extra, Transform, Load)

ETL is a process for extracting data from sources, transforming it into a strandardized format and loading it into a target system

ETL Steps:

Extract: Pull raw data from various source systems (operational databases, logs, external feeds) into a staging area. This step focuses on colleting data from diverse formats and sources.

Transform: Clean, normalise, remove duplicates, combine fields, enrich and apply business rules to ensure data quality and consistency; resolves schema mismatches and harmonises data for analytics.

Load: Move the transformed data into the target analytics store (data warehouse or data mart), enabling efficient querying and analysis. ETL is also common where transformation occurs after loading in target systems.

Why ETL is important

ETL enables integration of diverse data sources into a unified, governed analytic repository, supporting reliable reporting, governance and scalable analytics. It forms the foundation for classic data warehouse and modern BI workflows

Example
A bank pulls transaction data from ATMs, card payments and mobile Apps = it changes all dates to same format and flags fraud = loads it into the Fraud Data Mart

Question 10

Data Storage Formats

(1) Text-based Formats
- human readabe and easy to open

1.1 CSV (Comma-separated Values)
Simple taples, like Excel. Great for small data reports and straight forward interchange. Limitations include lack of native schema, no compression and poor handling of nested data.

1.2 JSON (JavaScript Object Notation)
Uses key value pairs. Flexible, common for APIs and MongoDB. Not optimal for large scale columnar analytics.

1.3 XML (Extensible Markup Language)
Uses tags like <name> Thaba </name>
It is older, but still used for configs and data exchange.

2. Big data / Analytics formats

Optomised for speed and storage.

2.1 Parquet

Column based compressed. Super fast for analytics and BI tools. Used in data lakes / warehouses. Due to fast columnar access and reduced Input/Output, it excels in large scale workloads on Hadoop / Spark ecosystems.

2.2 ORC (Optimized Row Columnar)

It is Similar to Parquet, It is popular in Hadoop / Hive.

2.3 Avro

It is row based, includes schema with the data.. it is often used with Kafka or similar streaming systems, where evolving schemas are important.

When to Use.

Best Format	Use Case
CSV	quick data exchange, simple datasets spreadsheets
JSON	Semi structured data from web services & logs, API's
Parquet	big analytical queries, columnar acess, performance storage efficiency
Avro	real time data streaming, robust schema evolution
XML	Legacy systems / configs

Section B: Artificial Intelligence and Machine Learning

Question 11

Artificial Intelligence (AI)

Definition

AI is the field concerned with creating machines or software that can perform tasks that would require intelligence when performed by humans. It is machines or software mimicking human cognative functions like learning, problem solving, perception, decision making, language and reasoning.

Narrow AI (Weak AI)

Narrow AI or Weak AI, referes to AI systems designed to perform a specific task or operate withing a limited domain. The systems do not possess general understanding or flexable, cross-domain reasoning. It follows patterns for that one specific job, and cannot learn to do unrelated tasks on its own.

Examples: Google Maps finding the fastest route. Siri's voice assistand. Both perform narrow task specific to programmed domanes

General AI (Strong AI)

General AI, also known as AGI - Artificial General Intelligence or Strong AI, refers to not yet achieved level of AI that would match or exceed human intelligence across a range of cognitive tasks. This would include planning, learning, reasoning and adapting to new domains.

Example: A machine that understands and responds like a human in any context. This has not been achieved as yet.

Question 12

Machine Learning (ML)

What is Machine Learning and How it differs from Traditional programming

Machine Learning (ML) is a subset of AI where systems learn from data without being explicitly programmed for every scenario. Machine learning systems improve with more data as it adapts as new data is distributed.

Traditional programming encodes explicit rules and logic to produce outputs. It follows hard coded, hand crafted rules

Three Main Types of ML

① Supervised Learning: Train the model with data that already has the right answers labeled

Example: Show it 1000 emails and tag specific emails as "spam" or "not spam".

It learns to filter emails as they come in.

② Unsupervised Learning: Give the mode data with no labels and it finds patterns on its own
Example: Group customers by buying habits without letting it what the groups are = it finds "budget shoppers" vs "luxury buyer".

③ Reinforcement Learning: The model learns by trial and error, getting rewards or penalties for actions.
Example: AI learns to play a game by getting 10 points for winning or -10 points for losing.

Question B

Deep Learing

Deep learning is a subset of machine learning that uses artificial neural networks with many layers, called "deep" neural networks. They are used to find patterns in huge amounts of data. It is inspired by how neurons connect in the human brain. Each layer learns more complex features than the last.

Key points

- Needs lots of data: Gets better with more examples.
- Learns features automatically: No need to

tell it what to look for, it figures it out.
- Great for messy data: Images audio, text, video.

Examples: Voice assistants understanding speech, Face unlock on phones. Self driving cars recognising stop signs.

	Relationship between AI, ML & DL		
	AI	ML	DL
What is it	Broad Idea: making machines act smart	A way to achieve AI: machines learn from data	A specific way to do ML: using deep neural networks
goal	mimic human intelligence for any task.	let systems improve at a task without explicit rules	Solve complex problems using layers of neurons
example	A chess robot, spam Filter ChatGPT	spam filter that learns from labelled emails	ChatGPT understanding and writing language
How it Works	Rules, logic, search or learn learning	Train on data to find patterns	Train huge neural nets on tons of data

All Deep Learning is Machine Learning. All Machine Learning is AI. Not all AI uses Machine Learning, and not all Machine Learning uses deep learning.

Example: A calculator programmed with "if" statements is AI, but not ML. A spam Filter trained on emails is ML, but not DL. Face unlock on your phone is DL.

Neural Networks: Neural networks are a family of ML models inspired by biological neurons. They consits of layers of interconnected units (neurons/nodes) with weights that can learn from data. They apply liner math operations, then run the results through non liner functions. This lets them reduce errors when training on labeled examples or find patterns in data when no labels or rewards are available

Question 14

Natural Language Processing

Natural Language Processing (NLP) is a field of study that studies the interaction between computers and human language. It focuses on enabling machines to understand, interpret, generate and respond to human language in a way that is both meaningful and useful.

3 Examples

- Language translation apps like Google Translate
  ↳ They translate text or speech between languages
- Chatbots for customer support
  ↳ they understand what client types through

25

natural language Understanding (NLU), form dialogue management by deciding what to do next, then finally generating a reply in human language.

- Voice assistants and speech recognition like Google Assistant or Siri / Alexa, trascribing and interpreting spoken language

Question 15

Large Language Model (LLM)

Large Language Model (LLM) is a type of ML model that specialises in natural language, trained on massive amounts of text so it can understand and generate human language. Large refers to terabytes of training data (books, websites and code) and billion of parameters.

LLMs are built to write text that makes sense and fits the context. They handle new tasks with just a few examples or none at all, and work across many language problems without needing a custom setup for each one. Compared to older NLP models, they are much bigger in both data and size, which gives them broader language skills. Traditional ML models are usually narrow and task specific, so they donot generalise as well.

2 Well Known LLMs

- GPT-4 / GPT-5 by OpenAI: This is one of the most widely used LLMs. It powers ChatGPT. It is used for writing and editing essays, emails, report. It is used for coding; debugging, generating code and explaining errors. Other uses are reasoning (solving problems, analysing data) and multimodel tasks (can also work with images and voice).

- Claude by Anthropic: This LLM is designed with a strong focus on safety and long context. It is mainly used for Long documents (summarising hundreds of PDFs, contracts, research papers), writing (business documents, technical writing), coding (software development and code review) and Safe dialogue (mostly used by companies where careful responses matter)

Question 16

Generative AI and How It Works

Generative AI systems produce novel data samples - text, images, audio - by learning data disributions from large datasets and then sampling or transforming those disributions. They map prompts to outputs using models such as LLMs for text or diffusion models for images. Training usually optimises likelihood or other generative objectives, and generation relies on learned representations to produce human like content.

3 Tools that Use Generative AI and What They Do

1. Chatbot Assistants and Writing Copilots:
They use large language modes to generate text, summarize documents, and help edit writing.
Example: ChatGPT and similar tools.

2. Image Generation Tools: Use diffusion modes to create new images from text prompts or Sketches. They support art, design work and synthetic media creation
Example: DALL·3 - OpenAI's model, now in ChatGPT. Generates images and follows complex prompts well.

3. Code generation and software assistants:
Use LLMs trained on code to write new code, autocomplete functions, explain existing code, and help debug errors.
Example: Github Copilot - Autocompletes code in your editor, explains code, writes tests. Built on Open AI's models.

28

Question 17

Training Dataset: Definition and Importance

A training dataset is the labeled or unlabeled data used to train an AI model by providing examples from which the model learns patterns, representations and mappings. High quality data is crucial for robust, accurate and generalisable models. Poor quality data leads to biased, unreliable and brittle models.

Risks of biased and Incorrect Data

Biased training data can reinforce or worsen discrimination, lead to unfair results and push models toward flawed or harmful behaviour. Issues like poor data quality, misleading or samples that don't represent the real world can hurt performance, limit how well a model generalizes and reduces trust in AI systems.

Possible harms include unfair or discriminatory decisions, privacy and ethical problems as well as wrong conclusions that cause bad real world outcomes.

29

Question 18

Computer Vision : what is it?

Computer vision is a field focused on enabling machines to enterpret and understand visual data from the world. It typically extracts features, objects, scenes and semantics from images or videos using algorithms, object detectors, segmentation models and related techniques.

How Machines See

They process images by converting pixels into ~~numerica~~ numerical representations, then applying learned models to detect patterns, recognise objects or infer scenes. Deep learning models, especially convolutional neural networks (such as CNN) and vision transformers, learn hierarchial features to classify or localize content in images and videos

Examples

- self driving cars detecting objects and navigating
- facial recognition for security or tagging photos

SECTION C: APIs, TOOLS and Developer Concepts

Question 19

Application Programming Interface (API)

Definition and Real world Analogy

An API is defined as a set of rules and protocols that enable one software component to request and exchange data or services from another. It is like a waiter in a resturant: you (the client) tell the waiter what you want (a data or service request), the waiter communicates with the kitchen (the service provider) using a fixed menu (the API). The waiter then returns your prepared dish (the response) back to you. This abstraction lets you get what you need without knowing how the kitchen prepares it.

APIs are primary interface for consuming IaaS, PaaS and SaaS offerings, enabling automated provisioning, management and integration across services and providers.

Importance in modern software and data systems

APIs let different systems work together like building blocks. Teams can connect services from multiple providers to build complex workflows without staring from Scratch. This setup supports automation, scales easily, and works across multicloud or hybid environments

For developers :
APIs give quick access to outside tools like login systems, databases or ML models. That means less reinventing, faster delivery and ability to build ecosystems with third party integrations.

For data
APIs are key for reading, adding or updating data automatically. They let systems, partnes and analytics pipelines share data in real time or near real time.

Examples
- API marketplace : like Rapid AI list thousands of APIs for things like weather, finance and translation. Developers can find, test and monitor them to add features to apps quickly.

- AI Model access : Companies like Anthropic and Open AI offer API endpoints for their language models. This lets developers build chatbots, summarise and coding assistants without running huge models themselves. It is a key example of APIs making AI scalable.

- Business workflows : An e-commerce site might use Stripe's API for payments, shipping

API's for delivery and CRM API for customer data. Each service connects through stable API contracts, so teams can work independently.

Question 20

API Key: Definition and Purpose

An API key is a credential used to identify and authenticate a calling program to an API. It acts like a password or token that proves the caller has permission to access the API and may be used for rate limiting, usage tracking and access control.

Storage and Protection best Practices

Never put API keys in client-side code like web or mobile apps or push them to public repositories. Store them server side using environment variables or a secrets manager. Rotate keys often and give each key only the minimum permissions it needs.

Where possible, use short lived tokens instead of long lived keys. For sensitive systems, use stronger controls lik OAuth or mutual TLS. Track API usage to spot weird activity and revoke compromised keys immediately.

Risks if Exposed Publicly

Leaked keys can be abused for unauthorised access, service overload or racking up charges on paid APIs. Attackers might also use the permissions to steal data or run credential-stuffing attacks. If a key leaks, one needs to rotate it fast and check access logs for damage.

Examples

- A developer accidentally commits an API key to public GitHub repo. Attackers find it and use the service, causing unexpected bills or data theft until the key is revoked
- A SaaS app integrates with a payment provider via API keys. Weak protection lets attackers hit the payment API, potentially draining test accounts or accessing customer data.

Question 21

REST API

Definition

REST API stands for Representational State Transfer. It is a way software systems talk to each other over the internet using standard web rules. It is a guide for building APIs.

How it Works

① Uses HTTP - same methods your browser uses:
GET read, POST create, PUT update, DELETE remove

② Stateless - each request has all the information needed. The server doesn't remember you between calls

③ Resources + URLs - Everything is "resource" with its own URL, Ex: api.com/users/123

④ JSON format - Data usually goes back and forth as JSON

What is it Used for

REST is used for Mobile apps talking to backends, Stripe payments, Google Maps data, Twitter feeds

example:

- A weather API: GET /current?location=x to fetch current conditions; POST used for submitting user generated weather reports in some designs; PUT/DELETE for updating or removing user submitted data in more advanced APIs.
- A RESTful API for a to-do app: GET /tasks to list tasks; POST / tasks to create a new task; PUT / tasks / {id} to update a task; DELETE/ tasks/ {id} to remove a task.

Main HTTP methods and Brief Descriptions

- GET: Retrieve a representation of a resource without side effects (read only). It should be safe and idempotent, used for fetching data.

POST: create a new resource or perform an action that changes state on the server. It is not idempotent by default and may return the created resource or status.

PUT: Update or replace a resource at a known URI. Idempotent: repeated requests yield the same result, Often used for full updates.

DELETE: Remove a resource at a given URI. Idempotent: deleting a non-existent resource typically yields a noop result or a not found result.

summary
GET – retrieve data (fetch user info)
POST – create new data (submitting a form)
PUT – update existing data (editing a profile)
DELETE – remove data (deleting an account)

Question 22

JavaScript Object Notation

Definition of JSON

JavaScript Object Notation (JSON) is a lightweight text format for storing and sending data. Uses key-value pairs and lists. it is humand and machine readable.

Why its commonly used with APIs & Data Exchange

- light weight : it is less bulky than XML, so faster to send over the internet

- Human readable : It is easy to debug and understand at a glance

- Language agnostic : every major programming language can read/writ JSON.

- Maps to code : Structure matches objects/dictionaries in code, so parsing is simple.

- Web-native : Browsers and HTTP work well with it, so it is the default of REST APIs

Simple example of a JSON object.

<> Code

```
{   "user_id" : 123,
    "name" : "Thabo",
    "email" : "thabo@gmail.com",
    "is_active" : true,
    "roles" : ["admin", "editor"]   }
```

The above shows strings, a number, a boolean and a list, which are all common JSON types.

Question 23

Prompt Engineering

Definition

it is the practice of crafting inputs (prompts) to AI models to elicit accurate, useful and controllable outputs. It is the skill of writing clear, specific instructions for AI models so they give you useful, accurate out put.

Why it matters

it is an important skill for AI tools like ChatGPT or Claude as

① Garbage in, garbage out; vague prompts get vague or wrong answers

② Saves time: Good prompts get you usable results on the 1st try

③ Unlocks capabilities; specific structure, examples, and constraints help the model reason, format and stay on topic

④ reduces hallucinations; giving clear context and instructions keep the AI grounded.

Examples of Prompts

①Poor : Write about dogs

(It is too vague. What kind or writing? For

38

who? How long?

Improved: Write a 150 word introduction for a grade 7. Science blog explaing why dogs have such a strong sense of smell. Use simple language and include 1 fun fact.

The above gives role, audiance length, tone and content focus

2) Poor: Summarise this
(no data provided, no length, no goa)

Improved: Summarise the article I will paste below into 3 bullet points. Focus only on the main causes of load shedding mentioned. Keep each bullet under 20 words.

The above defines, format, scope and constraints before you add the text.

3) Poor: "Write code for data analysis.
Improved: Write a Python function that loads a CSV with column: 'date', 'sales', 'region', cleans missing value and returns a pandas Data Frame with a 30 day moving average of sales grouping by region. Include error handling and a short usage example'

(39)

Question 24

MAP function Definition

A MAP function is a function that takes a list/array of items and a function, then applies that function to every item and returns a new list/array with the results. It transforms data 1-to-1

What it does in programming and data processing

① Transforms elements: Runs the same operation on each item without writing a loop.

② Keeps structure: Input of 10 items → output of 10 items, that are changed

③ Clean & declarative: You say what you want to do, not how to loop

④ Common in data work: Used for cleaning data, converting types, extracting fields, feature engineering

Examples

Python

<> Code

```
# Convert a list of Celsius temps to Fahrenheit
celsius = [0, 20, 30, 100]
fahrenheit = list(map(lambda c: c * 9/5 +
 32, celcius))
 print(fahrenheit)
 # Output: [32.0, 68.0, 86.0, 212.0]

# Capitalize names
name = ["thabo", 'sarah', "alex"]
capitalised = list(map(str.title.names))
print(capitalised)
# Output: ['Thabo', 'Sarah', 'Alex']
```

JavaScript

<> Code

```
// Square all numbers in an arry
const numbers = [1,2,3,4];
const squares = numbers.map(n=>n*n);
console.log(squares);
// Output: [1, 4, 9, 16]

// Extract just the emails from user objects
const users = [
  {name "Thabo", email: "thabo@example.com"},
```

```
{ name: "Sarah", email: "sarah@example.com" }
];
const emails = users.map(user => user.email);
console.log(emails);
// Output: ['thabo@example.com', 'sarah@example.com']
```

The key point: 'map' doesn't change the original list. It gives you a new one with transformed values.

Question 25

Cloud computing

Definition

Cloud computing is a model that uses servers storage, databases and software over the internet instead of owning physical hardware. You rent computing power on demand and pay for what you use. Like electricity from a grid vs buying you own generator

3 Main Cloud Service Models

① IaaS (Infrastructure as a Service)
Provides virtualised computing resources for customers to run their own software stack. For example VMs, storage, networking - computing resources. IaaS model manages

servers storage networking and virtualsation. For example: Amazon EC2 (AWS) or Microsoft Azure virtual machines.

② PaaS (Platform as a Service)

Offers a platform for customers to develop, deploy and run applications with out managing the underlying infrastructure. Example: Google App Engine, Microsoft Azure App Service.

③ SaaS (Software as a Service)

Delivers ready to use software applications over the internet, typically on subscription model!
Example: Google workspace, Salesforce, Microsoft 365.

The 3 models simplified

① IaaS: is like renting an empty apartment. You bring furniture and set it up
② PaaS: Is like renting a furnished apartment. Just move in and decorate
③ SaaS: Hotel room that is just used without a need to maintain it

In depth example of IaaS, PaaS, SaaS

① IaaS: Amazon EC2: Spin up virtual services where you install any OS/software you want. You manage the server.

② PaaS : Heroku - Push you code and it handles servers, scaling and deployment for you. You manage the app
③ SaaS : Google Workspace / Gmail - Full email app ready to use in your browser. You don't manage any infrastructure or code.

Question 26

Version Control System (VCS)

Definition and Importance

A Version Control System (VCS) tracks changes to files over time. It saves a history of every edit so that changes can be seen, who made the changes, when and roll back if needed.

What is Git and why it's used by data anallysts and Developers

Git is a distributed version control system also known as distributed VCS due to its speed, branching capabilities and strong collaboration work flow (pull requests, merges). Distributed means every person has the full project history on their machine, not just a central server

Why it is widely used

① Collaboration

Multiple people can work on the same code /notebooks without overwriting each other

② safety net -

Easy to undo mistakes or recover old versions

③ Branching

Test ideas or new features without breaking the main project

④ Industry standard

Works with GitHub, GitLab, Bitbucket. Most dev/ data jobs expect it.

⑤ For data analysts

Tracks changes to SQL scripts, Jupyter notebooks, dbt models and analysis code. Makes work reproducible.

Term	What is it	Simple analogy
Commit	A saved snapshot of your project at one point in time, with a message describing what changed	It is like hitting "save" in a video game. Creating a checkpoint one can return to
Branch	A seperate line of development copied from main code. Lets one work on stuff in isolation.	It is like a photocopy of an essay. Edits can be made on the copy without ruining the original copy

You make changes → commit to save them → create a branch to experiment → merge back when ready.

Question 27

Jupyter Notebook

Definition and Common Use

A Jupyter Notebook is an interactive environment or open source web app that combines executable code, rich text, visualisations and narrative in a single document. It supports many languages, but is mostly used with Python (via the IPython kernel) for data analysis, visualisation and exploration.

Popularity reasons

Data analysts and Data Scientists prefer it due to:

- reproducibility and literate programming: notebooks capture code, results and explanation in one place
- Ease of sharing: notebooks can be exported to HTML, PDF or shared in repositories and platforms like GitHub or Binder enabling collaboration and storytelling with data. Others can see code + results + narrative.

46

- interactive data exploration: immediate feedback, plots and inline documentation streamline experimentation. Great of EDA as data can be manipulated in one place and document insights can be viewed without switching tools

Question 28

Python definition & Why It is Popular

Python is a high leve, general purpose programming language known for simple, readable syntax that looks close to plain English.
It has a large ecosystem of libraries for data manipulation, analysis / visualisation and machine learning, and a strong community of enterprise adoptation. It integrates well with data workflows, notebooks and AI tooling. This makes it a default choice for data analysts and AI practitioners.

4 Commonly Used Python Libraries

① Panda
It handles tabular data with DataFrames eg excel in code. People use it to load CSVs, clean messy data, filter rows, groups join tables. The workhorse for data wrangling

② NumPy
Core numerical computing library providing

efficient multi-dimensional array operations and mathamatical functions, underpins most data science stacks.
Powers calculations behind the scenes. pandas is built on it. Used for numerical ops, statistics, linear algebra.

③ Scikit-learn:
Machine learning library with a wide range of algorithms (regression, classification, clustering) and utilities for preprocessing, model selection and evaluation.

④ Matplotlib / seaborn:
Makes basic charts on plots (matplotlib) and statistical plots built on top of matplotlib (seaborn). People use matplotlib for line charts, bar charts, histograms and scatter plots; giving full control to visualisation data. People use seaborn to make charts visually appealing with less code. it is great for heatmaps, box plots and exploring relationships.

48

SECTION D: Data Ethics, Governance & Careers

Question 29

Data Privacy and Importance

Data privacy is the protection of individual' personal information from unauthorised access, use, disclosure or manipulation, and the protection of individuals' control over their own data. This also includes how data is collected, stored, processed and shared. It is important to practise good governance in data collection as it builds trust, ensures compliance and protects individuals' rights.

General Data Protection Regulation (GDPR)

The GDPR is a comprehensive regulation enacted by the European Union to harmonise data protection laws across EU member States and to give individuals strong rights over their personal data. This includes trasparency, access, correction, deletion and data portability with a framework for lawful processing, consent, and accountability.
It imposes obligation on organisations that process personal data of individuals located in the EU, regardless of the organisation

location, and it applies to data controllers and processors engaged in EU data activities

Scope and Geography: GDPR applies to:

- All organisations that process the personal data of individuals in the EU, regardless of the organisation's location
- Processing activities include collecting, storing, using, sharing or deleting personal data and include special catagories of data and profiling activities that require heigtened protections
- It also applies to organsations outside the EU if they offer goods or services to, or monitor the behaviour of individuals within the EU (eg targeting or tracking)
- Key rights under GDPR include access to data, correction, erasure (right to be forgotten) restriction of processing, data portability, objection and consent withdrawal; along with duties such as data minimisation, purpose limitation and ensuring data security.

- Consequences of non compliance can include substantial fines, corrective action orders and reputational damage. GDPR emphasizes accountability, documentation and assessments for high risk processing.
- The GDPR's global influence is widely

recognised in policy and practice discussions around privacy, data governance and corporate compliance; including its use as a benchmark in privacy-by-design and risk management practices in many jurisdictions and organisations worldwide.

Question 30

Data Ethics Definition

Data ethics is the rules and principals for collecting, using and sharing data responsibly. It asks the question: "Even if we can use this data, should we?". It focuses on privacy, consent, fairness, transparency and preventing harm

Examples of Unethical Use of data or AI

① Biased hiring algorithm

Discriminatory algorithmic decision-making in hiring. If data driven hiring systems are trained favour and disfavour certain groups of candidates (eg favour males over females), then it is unethical.

Reason: This is unethical because it violates principles of fairness, equal opportunity and non-discrimination and can entrench social inequities.

② Selling location data from apps

Data usage beyond stated purpose (purpose creep) or opaque data practices as well as selling location history to ad brokers, where uses dont realise that they are being tracked.

Reason : it breaches consent based norms and can lead to misuse or exploitation of personal information. For example individuals can be exposed, by their sensitive information, such as home address, falling in the wrong hands.

③ Surveillance based interdiction without consent. Systems that increasingly monitor individual' behaviour (eg. in workplace or public spaces) to infer sensitive attributes or predict behaviour without explicit consent raises privacy harms and autonomy concerns.

Reason ; This unethical as it is an infringement of privacy rights and potential chilling effects, especially when used to justify punitive or discriminatory practices.

Question 31

Data Governance Definition

Data governance is the system of decision rights and responsibilities for information mangement. It is to ensure data quality, availability, usability, integrity and security across an organisation. It encompasses policies, processes, standards and metrics to manage dat as a strategic asset.

Key Components of Good Data Governance Framework

① Scope and objectives:
Defines data domains, ownership and intended outcomes aligned with business strategy.

② Data ownership and stewardship:
Assign data owners, stewards and roles with clear accountability for data assets and lifecyle management. Prevents chaos and finger pointing.

③ Data quality management:
Standards, profiling, cleansing, validation and remediation process to ensure accuracy, completeness and consistency.

④ Data policies and Standards:
Formal rules governing data defintions, metadata, lineage, privacy, security, retention and usage - for example alignment to laws like POPIA / GDPR. Aids in avoiding fines, breaches and reputation damage.

5. <u>Data lifecyle and Metadata Management</u>: Classification, cataloging, lineage tracing, and contextualisation of data assets (metadata) for discoverability and governance. This is basically a "library card" system for data. It aids in avoiding duplications as well.

6. <u>Data Security and Privacy Controls</u>: Access management, encryption, audit trails and compliance with laws such as GDPR; risk based controls and data minimisation practices. Prevents leaks and misuse of data.

7. <u>Oversight and governance bodies</u>: Steering committees, data governance councils and defined decision processes for escalations and approvals.

8. <u>Measurement and Continous Improvements</u>: KPIs and metrics for data quality, usage, compliance and ~~governe~~ governance maturity, with ongoing program refinement.

Good data governance programs integrates people, processes and technology. It includes data catalogs, metadata management tools and governance enabled data platforms to operationalise governance goals.

Question 32

Bias AI definition

Bias AI refers to systematic errors or prejudices in data, models that causes unfair, inaccurate, or skewed results for a certain group or situation. The model learned patterns from data that don't reflect reality fairly.

What it means for AI model to be biased

- It doesn't treat groups equally:
  Performance is worse for some genders, races, ages, regions.

- It learned bias from data:
  If training data reflects historical discrimination, the mode copies it.

- it amplifies problems:
  Small bias in data becomes large scale automated decisions

- It can be hidden:
  The model seems objective because it's math, but the outcomes aren't fair.

Other known cases of AI bias
- Facial recognition may have a higher error rate for darker skin tones, recruiting tools may downrank

women and individuals who are not caucasian, loan models often penalise certain area codes.

Real word example of AI Bias and Consequences:

- Example : Health Risk algorithm in Hospitals (USA)
- What happened: The algorithm used on 200million? patients to identify who needed extra care used "health care cost" as a proxy for "health need". Black patients get charged less historically due to access barriers, so the model scored them as healthier than equally sick white patients
- Potential consequences: ① Denied care - Black patients were less likely to be referred to programs for complex care needs. ② Widened healthgaps - existing dis-parities got automated and scaled. ③ Loss of trust - Patients and doctors lose faith in AI tools meant to help.

The above model wasnot intentionally racist, but using cost as a proxy for health baked in decades of unequal acces. The bias was in the data and design choice.

Question 33

Role	Main Job	Core Skills	Common Tools
Data Analyst	Focuses on data querying, reporting, visualisation and business insights	SQL, Excel, data visualisation, basic stats, business acumen, storytelling	SQL, Excel / Google Sheets, Power BI, Tableau, Python (pandas), Jupyter
Data Scientist	Predict the future & finds patterns	Emphasised statistical modeling, machine learning, Python / R, hypothesis testing, feature engineering, experimentation	Python, R, scikit-learn, Jupyter, SQL, statsmodels, seaborn, Git
Data Engineer	Builds pipelines to move / clean / store data.	SQL, ETL/ELT, data modeling, cloud, distributed systems, APIs, data warehousing	Python, SQL, Spark, Airflow, dbt, Snowflake, BigQuery, Kafka, Git
AI Engineer	Put ML/AI models into production	MLOps, model deployment, APIs, software engineering, cloud, LLMs, vector DBs	Python, Docker, FastAPI, PyTorch / TensorFlow, LangChain, AWS / GCP, MLFlow, Git

Question 34

Business Intelligence (BI)

Business Intelligence (BI) is the process, methodologies and technologies used to collect, transform, analyse an present data to support business decisions and decision making. It emphasises turning data into actionable insights through dashboards and performance tracking.

Popular BI Tools

① Power BI.

It is a microsoft BI platform enabling data visualisation, dashboards and reporting. it helps organisations consolidate data sources, create interactive reports and share insights across an organisation, supporting data driven decision makings.

② Tableau:

A data visualisation and analytics platform that allows users to build interactive dashboards, explore data intuitively and communicate insights effectively to stakeholders, facilitating data driven decision's

How BI tools help businesses

① They provide self service analytics and dashboards that democratise access to data, enabling quicker, informed decisions across departments.

② They enable monitoring of key performance indicators (KPIs) and trends, supporting scenario analysis and performance management (eg sales, operations and customer behavior).

SECTION E: Personal Reflection

Question 35

Interesting Concepts

SQL, JSON, Javascript and Python. Have often heard the terms being mentioned on social media (TikTok & X). It was an eye opener to learn what each really entail and how they fit in our daily lives.

SQL: Teaches how to be logical and filter information step by step
Python: How to break big problems into small ones
JSON: How to organise information clearly

Javascript: The cause-effect, basically what happens when a user ~~ct~~ clicks

Question 36

Difficult Concepts

SQL, JSON, Java Script and Python, basically all coding tools

The main reason I enrolled for the short course was to be in a classroom environment where i can interact/ask questions if I don't understand. Platforms like YouTube and the internet will also help me better understand concepts. I learn better in an interactive environment.

Question 37

Effects of Data and AI on career

I have an honours degree in Business Management and currently busy studying a degree in Logistics and Operations Management. I feel that AI and data have a great impact on my future aspiration of being a Business Manager or Logistics Manager as both replace gut feel and manual work. They assist in providing accurate insights to make it easier to focus on decisions.

3 big wins

- Prediction: will be able to spot churn, stockouts or late deliveries before they happen
- Optimise: AI finds better prices, routes and stock levels humans miss
- Automate: kills repeatitive copy and paste work. Repons, trackirs and alerts run them-selves.

Skills I will learn will set me appart for career oprunities in future, basically giving me an edge over my competiters.

References

Section A

1. Chou, D., Tripuramally, H., & Chou, A. (2005). BI and ERP integration. Information Management & Computer Security, 13(5), 340-349. https://doi.org/10.1108/09685220510627241

2. Counihan, A., Finnegan, P. & Sammon, D. (2002). Towards a framework for evaluating investment in data warehousing. Information Systems Journal, 12(4), 321-338. https://doi.org/10.1046/j.1365-2575.2002.0013.x

3. Dmitriyev, V., Mahmoud, T., & Marín-Ortega, P. (2002) SOA enabled ELTA: Approach in Designing Business Intelligence Solutions in Era of Big Data. International Journal of Information Systems and Project Management, 3(3), 49-63. http://doi.org/10.12821/ijispm030303

4. Fortino, G., Guerrieri, A., Pace, P., Savaglio, C. & Spezzano, G (2022) IOT Platforms and Security: An Analysis of Leading Industrial/Commercial solutions. Sensors, 22(6), 2196 https://doi.org/10.3390/s22062196

5. Gozali, A., Romadhony, A., & Subaveerapandiyan, A. (2023), One Data Indonesia Policy Adaptation for Telkom University Data Ware-

house Framework, Register Jurnal Ilmiah Teknologi Sistem Informasi, 9(2), 160-176
https://doi.org/10.26594/register.v9i2.3473

6. Han, Y. (2013) IaaS Cloud Computing Services for Libraries: Cloud Storage and virtual machines. Oclc Systems & Services, 29(2), 87-100
https://doi.org/10.1108/10650751311319296

7. Kelly, D., Glavin, F., & Barret, E. (2020). Serverless Computing: Behind the Scenes of Major Platforms 304-312.
https://doi.org/10.1145/3014812.3014829

8. Niedrite, L., Solodovnikova, D., & Grundman L. Development of Data Warehouse Conceptual Models., 1-23
https://doi.org/10.4018/978-1-60566-232-9.ch001

9. Odiths, I., Bicevska, Z., Bicevskis, J., & Karniths, G. (2018), Implimentation of NoSQL Based Data Warehouses. Baltic Journal of Modern Computing, 6(1). https//doi.org/10.22364/bjmc.2018.6.1.04

10. Pedrozo, W. and Periera, L (2023) Building a Data Webhouse to support Click stream Data Analysis. Journal of Engineering Reaserch, 3(40), 2-11.
https://doi.org/10.22533/at.ed.317340232383116

11. Plase, D (2017), A Systematic Review of SQL-On-Hadoop by Using Compact Data Formats. Baltic Journal of Modern Computing, 5(2)
https://doi.org/10.22364/bjmc.2017.5.2.06

12. Ramesh, K. (2024). Efficient Security for Resource-Constrained Devices with LAN: A lightweight Protocol Utilizing Chaotic Map-Based S-Box.
https://doi.org/10.22541/au.170664330.05586375/v1

13. Scheible, R., Thomczyk, F., Blum, M., Rautenberg, M., Prunotto, A., Yazijy S., ... & Boeker, M. (2023). Integrating row level security in i2b2: Segregation of Medical Records into Data Marts without data replication and synchronization. Jamia Open, 6(3).
https://doi.org/10.1093/jamiaopen/ooad068

14. Supriyandoko, F. (2022). Building Data Marts in Teaching Management: A case Study. Pupil International Journal of Teaching, Education and Learning, 6(3), 01-14.
https://doi.org/10.20319/pijtel.2022.63.0114

15. Visweswaran, S., McLay, B., Cappella, N., Morris, M., Milnes, J., Reis, S., ... & Becich, M. (2021). An atomic approach to the design and implementation of a research data warehouse. Journal of the American Medical Informatics Association, 29(4), 601-608.
https://doi.org/10.1093/jamia/ocab204

64

16. Watson, H. (2002). Recent Developments in Data Warehousing. Communications of the Association for Information Systems, 8.
https://doi.org/10.17705/1cais.00801

17. Watson, H., Wixom, B., Buonamici, J., & Revak, J. (2021). Sherwin-Williams' Data Mart Strategy: Creating Intelligence Across the Supply Chain. Communications of the Association for Information Systems, 5.
https://doi.org/10.17705/1cais.00509

SECTON B

1. Anand, N. (2024). Of Code and Consequences: Assessing the Impact of Artificial Intelligence on International Criminal Law Norms Governing the Direct and Public Incitement to Genocide. International Journal of Legal Information, 52(2), 166-175
https://doi.org/10.1017/jli.2024.39

2. Appleby, R. and Basran, P. (2022). Artificial Intelligence in veterinary medicine. Journal of the American Veterinary Medical Association, 260(8), 819-824.
https://doi.org/10.2460/javma.22.03.0093

3. Bansal, R. and Sangwan, A. (2024). A Comprehensive Review of Artificial Intelligence. International Journal of Multidisciplinary Research, 6(5).

https://doi.org/10.36948/ijfmr.2024.v06i05.28982

(4) Baaulience, V., Bilan. Y., Navickas, V., a Civin, L. (2023). The Aspects of Artificial Intelligence in Different Phases of the Food Value and Supply Chain. Foods, 12(8), 1654.
https://doi.org/10.3390/Foods12081654

(5) Cook, J and Cook, J. (2024). Artificial Intelligence in Management Education: Transformative Potential and Challanges. SAMANJ, 89(4), 340-355.
https://doi.org/10.1108/samamj-05-2024-0026

(6) Devedzic, V. (2020): Is this artificial intelligence?. Facta Universitatis - Series Electronics and Energetics, 33(4), 499-529.
https://doi.org/10.2298/Fuee2004499d

(7) Dignam, A (2020). Artificial Intelligence, Tech Corporat Governance and The Public Interest Regulatory Response. Cambridge Journal of Regions Economy and Society, 13(1), 37-54
https://doi.org/10.1093/cjres/rsaa002

(8) Etscheid, J. (2019). Artificial Intelligence in Public Administration., 248-261
https://doi.org/10.1007/978-3-030-27325-5_19

(9) Frank, D and Otterbring, T. (2023). Autonomy, Power and the Special Case of Scarcity: Consumer Adoption of Highly Autonomous Artificial

(66)

intelligence. British Journal of Management, 35(4), 1100-1123
https://doi.org/10.1111/1467-8551.12780

(10) Govia, L (2020), Coproduction, Ethics and Artificial Intelligence: A Perspective from Cultural ~~Anthropolo~~ Anthropology. Journal of Digital Social Research, 2(3), 42-64.
https://doi.org/10.33621/jdsr.v2i3.53

(11) Hanssen, A. and Nichele, S. (2019). Ethics of Artificial Intelligence Demarcations., 133-142
https://doi.org/10.1007/978-3-030-35664-4_13

(12) Kendale, S (2021). The role of Artificial intelligence in Pre Operative Medicine. International Anesthesiology Clinics, 60(1), 69-73.
https://doi.org/10.1097/aia.0000000000000344

(13) Laat, P. (2021). Companies Committed to Responsible AI: From Principles towards Implementation and Regulation?. Philosophy & Technology. 34(4), 1135-1193
https://doi.org/10.1007/s13347-021-00474-3

(14) Luchs, I., Apprich, C., & Broerma, M. (2023). Learning machine learning: On the political economy of big tech's online AI cources. Big Data & Society, 10(1),

https://doi.org/10.1177/20539517231153806

15 Marino, M., Vasquez, E., Dieker, L., Basham, J., & Blackorby, J (2023). The future of Artificial Intelligence in Special Education Technology. Journal of Special Education Technology, 38(3), 404-416.
https://doi.org/10.1177/01626434231165977

16 Molet, N. (2023). AI and Organisational Transformation: Antropological Insights into Higher Education, Izzivi Prihodnosti. 8(3, August 2023)
https://doi.org/10.37886/ip.2023.007

17 Spector, J. and Ma, S. (2019). Inquiry and critical thinking skills for the next generation: From Artificial Intelligence back to Human Intelligence: Smart Learning Environments, 6(1).
https://doi.org/10.1186/s40561-019-0088-z

18 Tedre, M., Toivonen, T., Kahila, J., Vartiainen H., Valtonen. T., Jormanainen, I., ... & Pears, A (2021). Teaching Machine Learning in K-12 Classroom: Pedagogical and Technological Trajectories for Artificial Intelligence Education: Ieee Access, 9, 110558 - 110572.
https://doi.org/10.1109/acess.2021.3097962

19 Vlist, F., Helmond, A, & Ferrari, F. (2024), Big AI: Cloud infrastructure dependence and Industrialisation

of Artificial Intelligence, Big Data & society 11(1)
https://doi.org10.1177/20539517241232630

SECTION C

1) Altemimi, M. and Alasadi, A (2022). Ecommerce based on Cloud Computing: The Art of State: European Journal of Information Technologies and Computer Science, 2(4), 7-13
https://doi.org/10.24018/compute.2022.2.4.56

2) Azadi, M, Toloo, M., Ramezani, F., Saen, R., Hussain, F. & Farnoudkia, H. (2023). Evaluating efficiency of Cloud Service Providers in Era of Digital Technologies. Annals of Operations Research, 342 (2), 1049-1078.
https://doi.org/10.1007/S10479-023-05257-x

3) Galvin, D and Sun, M. (2012). Avoiding the Death Zone: Choosing and Running the Library Project in the Cloud. Library Hi Tech, 30(3).418.427.
https://doi.org/10.1108/07378831211266564

4) Guo. R., Tafti, A & Subramanyam, R. (2023) Internal IT modularity, Firm, Size and Adaptation of Cloud Computing. Electronic Commerce Research, 25(1), 319-348
https://doi.org/10.1007/S10660-02309691-8

5) Hoefer, C. and Karagiannis. G. (2010). Taxonomy of Cloud Computing Services, 1345-1350

https://doi.org/10.1109/glocomw.2010.5700157

6. Hansen, J. Wessels, S & Eckert, C (2016). (Calcay.) 314-323
https://doi.org/10.1145/2991079.2991117

7. Hsu, I. and Cheng, F. (2013). SaaS. A cloud Computing Service Model using semantic based agent. Expert Systems, 32(1). 77-93
https://doi.org/10.1111/exsy.12063

8. Iqbal, S., Kiah, M., Anuar, N., Daghighi, B., Wahab, A., & Khan, S. (2016). Service delivery models of Cloud Computing: Security Issues and Open Challenges. Security and Communication Networks, 9(17), 4726-4750.
https://doi.org/10.1002/sec.1585

9. Jiang, P and Wang, W (2022). Comparison of SaaS and IaaS in cloud ERP implementation: The lessons from practitioners. Vine Journal of Information and Knowledge Management Systems. 54(3), 683-701.
https://doi.org/10.1108/vjikms-10-2021-0238

10. Mazumdar, A and Alharahsheh. H (2019). Insights of Trends and Developments in Cloud Computing. South Asian Research Journal of Engineering and Technology, 01(03). 98-107
https://doi.org/10.36346/sarjet.2019.v01i03.003

2

11) Miyachi. C. (2018). What is "cloud"? It is time to update the NIST definition? Iee cloud Computing, 5(3), 671
https://doi.org/10.1109/mcc.2018.032591611

12) Paraiso, F, Haderer, N., Merle, P., Rouvoy, R. a Seinturier, L. (2012). A Federated Multi-cloud Paas infrastructure., 392-399.
https://doi.org/10.1109/cloud.2012.79

13) Sala. Zarate, M. and Colombo. Mendoza. L. (2012). Cloud Computing: A Review of Paas, Iaas, Saas Services and Providers. Lampsakos. (7), 47.
https://doi.org/10.21501/21454086.844

14) Sutherland, S (2014). Secure APIs and Protocols to Connect Enterprise Applications of Cloud Services:
https://doi.org/10.28945/2020

15) Zhang. Y. and Huang. L, (2014). The Research of Cloud Computing Services Model. Applied Mechanics and Materials, 556-562, 6262-6265
https://doi.org/10.4028/www.scientific.net/amm.556-562.6262

SECTION D

1) Avnoon, N (2023), Replace, absorb, serve: Data Scientists Talk about their aspired jurisdictions -

(71)

dichon: Current Sociology, 72(6), 1107-1120

(2) Bie, T., Raedt, L., Hernandez-Orallo, J., Hoos, H., Smyth, P., and Williams, C. (2022) Automating Data Science. Communications of the ACM, 65(3), 76-87
https://doi.org/10.1145/3495256

(3) Cao, L. (2019) Data Science: Profession and Education IEEE Intelligent Systems, 34(5), 35-44.
https://doi.org/10.1109/mis.2019.2936705

(4) Davenport, T. (2020), Beyond Unicorns: Educating, classifying and Certifying Business Data Scientists. Harvard Data Science Review.
https://doi.org/10.1162/99608f92.55546b4a

(5) Eckert, C., Isaksson, O., Eckert C., Coeckelbergh, M., a Hagstrom, M. (2020) Data Fairy in Engineering Land: The Magic of Data Analysis as a Sociotechnical process in Engineering Companies Journal of Mechanical Design. 142 (12)
https://doi.org/10.1115/1.4047813

(6) Fayyad, C., Isaksson, O., Eckert, C., Coeckelbergh, M., a Hagstrom, M. (2020). Data Fairy in Engineering Land: The Magic of Data Analysis as a Sociotechnical Process in Engineering Companies. Journal of Mechanical Design, 142 (12)
https://doi.org/10.1162/99608f92.00865006

(7) Ferino, S. (2025). Walking the Tightrope of LLMs for Software Development: A Practitioners' Perspective. http://doi.org/10.48550/arxiv.2511.06428

(8) Johnson, M., Jain, R., Brennan-Tonetta, P., Swartz, E., Silver, D., Paolini, J., ... & Hill, C. (2021). Impact of Big Data and Artificial Intelligence on Industry: Developing a Workforce Roadmap for a Data Driven Economy. Global Journal of Flexible Systems Management, 22(3), 197-217. https://doi.org/10.1007/s40171-021-00272-y

(9) Mauro, A., Greco, M., Grimaldi, M., & Ritala, P. (2018). Human Resources for Big Data professions: A systematic classification of job roles and required skill sets. Information Processing & Management, 54(5), 807-817 https://doi.org/10.1016/j.ipm.2017.05.004

(10) Munteanu, I., Newcomer, K., & Best, C. (2024). Building an evidence engine to promote more responsive Government. Public Administration Review, 85(4), 947-961. https://doi.org/10.1111/puar.13880

(11) Oza, B. and Behnaz, A. (2024). Using Knowledge Graphs for Enabling Collaborative Financial Market Data Analytical Processes. https://doi.org/10.20944/preprints202407.1021.v1

(12) Quan, T. and Raheem, M. (2023). Human Resource

Analytics on Data Science Employment Based on Specialized Skill Sets with Salary Prediction. International Journal on Data Science, 4(1), 40-59. https://doi.org/10.18517/ijods.4.1.40-59.2023

8 Rantonen, M and Karjalainen, M (2022). Development of Data-Driven Curriculum Cources in the Bachelor of Engineering Degree Program. Journal of Strategic Innovation and Sustainability, 17(1) https://doi.org/10.33423/jsis.v17i1.5186